

**Department of Artificial Intelligence and Data Science****Vision of the Department**

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2025-2026

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

Krishna Mishra (18/08/2026)

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Session	2025-26 (ODD)	Course Name	LAB:Neural Networks and Deep Learning
Semester	7th	Course Code	23ADS1704
Roll No	41	Name of Student	Krishna Mishra

Practical Number	05
Course Outcome	CO1: Understand the fundamental concepts of deep learning and neural network architectures CO2: Apply different techniques to develop and optimize deep learning models. CO3: Analyse the performance of deep learning models using appropriate evaluation techniques. CO4: Design and implement deep learning solutions for real-world problems using modern tools and frameworks.
Aim	Implement Transfer Learning for Image Classification using pre-trained models.
Theory	Transfer learning in computer vision revolves around reusing structural knowledge acquired by a deep neural network on a massive benchmark dataset, such as ImageNet, and applying it to a distinct, target classification problem. When training a convolutional neural network or vision transformer from scratch, early layers require immense amounts of data and compute to learn foundational visual concepts like edges, textures, and color gradients. By adopting pre-trained weights instead of random initializations, transfer learning bypasses this resource-intensive initial phase. The model begins training already equipped with general pattern-recognition capabilities, which drastically speeds up convergence, reduces total computational requirements, and mitigates severe overfitting issues when working with limited target datasets.



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Procedure and Execution

• Data Preprocessing & Standardisation

- Load and split the target dataset into training, validation, and test sets.
- Resize target images to fit the pre-trained model's required input dimensions.
- Apply the exact normalization (pixel scaling, mean, and standard deviation subtraction) used during the backbone's original pre-training.

• Model Architecture Adaptation

- Load the pre-trained base model (e.g., ResNet, MobileNet) with weights trained on a large dataset (e.g., ImageNet).
- Remove the original pre-trained classification layer (top head).
- Attach a new classification head consisting of Global Average Pooling, Dropout (for regularization), and a Dense output layer matching the target number of classes.

• Phase 1: Feature Extraction (Training Classifier Head)

- **Freeze Backbone:** Set all weights in the pre-trained feature extractor to non-trainable ($\text{trainable} = \text{False}$).
- **Compile:** Configure the model using a standard learning rate (e.g., $\eta = 10^{-3}$) and a suitable loss function (e.g., categorical cross-entropy).
- **Train:** Pass normalized images through the frozen backbone, backpropagate gradients, and update **only** the parameters of the custom classification head until validation loss reaches a baseline stability.

• Phase 2: Fine-Tuning (Optional Optimization)

- **Unfreeze Select Layers:** Set the top convolutional layers of the pre-trained backbone to trainable ($\text{trainable} = \text{True}$).
- **Lock Batch Normalization:** Keep Batch Normalization layers locked in inference mode to prevent altering moving mean/variance statistics.



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- **Recompile with Lower Learning Rate:** Reduce the optimizer's learning rate drastically (e.g., $\eta = 10^{-5}$) to prevent destroying pre-learned representations (*catastrophic forgetting*).
- **Train to Convergence:** Resume training over both the unfrozen base layers and classification head until validation performance peaks.
- **Evaluation & Inference**
 - Evaluate the fine-tuned model on the unseen test dataset.
 - Deploy the finalized model for predictions on new target images.

Code:

```
#
=====
====
# FLOWER IMAGE CLASSIFICATION USING TRANSFER LEARNING
# MobileNetV2 Pre-trained on ImageNet
# Google Colab - Complete Code
#
=====
====

# -----
# 1. IMPORT LIBRARIES
# -----

import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
import os

from tensorflow.keras import layers, models
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau

print("TensorFlow Version:", tf.__version__)

# -----
# 2. DOWNLOAD FLOWER DATASET
# -----
```



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```
dataset_url =
"https://storage.googleapis.com/download.tensorflow.org/example_images/flower_photos.tgz"

data_dir = tf.keras.utils.get_file(
    "flower_photos",
    origin=dataset_url,
    untar=True
)

# Correct data_dir to point to the directory containing actual class subfolders
# The dataset usually unpacks into a nested 'flower_photos' directory
data_dir = os.path.join(data_dir, 'flower_photos')

print("\nDataset downloaded successfully!")
print("Dataset path:", data_dir)

# -----
# 3. PARAMETERS
# -----
IMG_SIZE = (224, 224)
BATCH_SIZE = 32
SEED = 123

# -----
# 4. LOAD DATASET
# -----
train_ds = tf.keras.utils.image_dataset_from_directory(
    data_dir,
    validation_split=0.2,
    subset="training",
    seed=SEED,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

validation_ds = tf.keras.utils.image_dataset_from_directory(
    data_dir,
    validation_split=0.2,
    subset="validation",
    seed=SEED,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

class_names = train_ds.class_names
num_classes = len(class_names)
```



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```
print("\nFlower Classes:")
for i, name in enumerate(class_names):
    print(i, ":", name)

print("\nNumber of classes:", num_classes)

# -----
# 5. DISPLAY SAMPLE IMAGES
# -----
plt.figure(figsize=(10, 8))

for images, labels in train_ds.take(1):
    for i in range(9):
        ax = plt.subplot(3, 3, i + 1)
        plt.imshow(images[i].numpy().astype("uint8"))
        plt.title(class_names[labels[i]])
        plt.axis("off")

plt.suptitle("Sample Flower Images", fontsize=16)
plt.show()

# -----
# 6. IMPROVE DATA PIPELINE PERFORMANCE
# -----
AUTOTUNE = tf.data.AUTOTUNE

train_ds = train_ds.prefetch(buffer_size=AUTOTUNE)
validation_ds = validation_ds.prefetch(buffer_size=AUTOTUNE)

# -----
# 7. DATA AUGMENTATION
# -----
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.15),
    layers.RandomZoom(0.15),
    layers.RandomContrast(0.1)
])

# -----
# 8. LOAD PRE-TRAINED MOBILENETV2
# -----
base_model = MobileNetV2(
    weights="imagenet",
    include_top=False,
    input_shape=(224, 224, 3))
```



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	<pre>) # Initially freeze all pretrained layers base_model.trainable = False print("\nMobileNetV2 loaded with ImageNet weights.") print("Pre-trained layers are initially frozen.") # ----- # 9. CREATE TRANSFER LEARNING MODEL # ----- inputs = layers.Input(shape=(224, 224, 3)) # Data augmentation x = data_augmentation(inputs) # MobileNetV2 preprocessing x = preprocess_input(x) # Extract features using MobileNetV2 x = base_model(x, training=False) # Convert feature maps to a single vector x = layers.GlobalAveragePooling2D()(x) # Dropout to reduce overfitting x = layers.Dropout(0.2)(x) # Flower classification layer outputs = layers.Dense(num_classes, activation="softmax")(x) model = models.Model(inputs, outputs) # ----- # 10. COMPILE MODEL # ----- model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001), loss="sparse_categorical_crossentropy", metrics=["accuracy"]) print("\nModel created successfully!")</pre>
--	--



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```
# -----
# 11. DISPLAY MODEL SUMMARY
# -----
model.summary()

# -----
# 12. CALLBACKS
# -----
early_stopping = EarlyStopping(
    monitor="val_loss",
    patience=3,
    restore_best_weights=True
)

reduce_lr = ReduceLROnPlateau(
    monitor="val_loss",
    factor=0.2,
    patience=2,
    min_lr=1e-7
)

# -----
# 13. TRAIN TRANSFER LEARNING MODEL
# -----
INITIAL_EPOCHS = 10

print("\n=====")
print("STARTING TRANSFER LEARNING")
print("=====")

history = model.fit(
    train_ds,
    validation_data=validation_ds,
    epochs=INITIAL_EPOCHS,
    callbacks=[early_stopping, reduce_lr]
)

# -----
# 14. FINE-TUNE MOBILENETV2
# -----
print("\n=====")
print("STARTING FINE-TUNING")
print("=====")

# Unfreeze MobileNetV2
base_model.trainable = True
```



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```
# Freeze the first 100 layers
for layer in base_model.layers[:100]:
    layer.trainable = False

# Recompile using a very small learning rate
model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)

FINE_TUNE_EPOCHS = 10

total_epochs = INITIAL_EPOCHS + FINE_TUNE_EPOCHS

history_fine = model.fit(
    train_ds,
    validation_data=validation_ds,
    epochs=total_epochs,
    initial_epoch=len(history.history["loss"]),
    callbacks=[early_stopping, reduce_lr]
)

# -----
# 15. COMBINE TRAINING HISTORY
# -----

train_accuracy = (
    history.history["accuracy"] +
    history_fine.history["accuracy"]
)

val_accuracy = (
    history.history["val_accuracy"] +
    history_fine.history["val_accuracy"]
)

train_loss = (
    history.history["loss"] +
    history_fine.history["loss"]
)

val_loss = (
    history.history["val_loss"] +
    history_fine.history["val_loss"]
)

# -----
# 16. EVALUATE MODEL
```

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```
# -----
print("\n=====")
print("FINAL MODEL EVALUATION")
print("=====")

test_loss, test_accuracy = model.evaluate(
    validation_ds,
    verbose=1
)

print("\n-----")
print("FINAL RESULTS")
print("-----")
print("Validation Loss   :", round(test_loss, 4))
print("Validation Accuracy :", round(test_accuracy * 100, 2), "%")
print("-----")

# -----
# 17. DISPLAY ACCURACY
# -----
final_train_accuracy = train_accuracy[-1] * 100
final_val_accuracy = val_accuracy[-1] * 100

print("\n=====")
print("ACCURACY")
print("=====")
print(f"Training Accuracy : {final_train_accuracy:.2f}%")
print(f"Validation Accuracy : {final_val_accuracy:.2f}%")
print(f"Final Model Accuracy: {test_accuracy * 100:.2f}%")
print("=====")

# -----
# 18. PLOT ACCURACY AND LOSS
# -----
plt.figure(figsize=(14, 5))

# Accuracy plot
plt.subplot(1, 2, 1)

plt.plot(
    train_accuracy,
    label="Training Accuracy",
    linewidth=2
)

plt.plot(
    val_accuracy,
    label="Validation Accuracy",
```



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	<pre>linewidth=2) plt.axvline(x=INITIAL_EPOCHS - 1, color="red", linestyle="--", label="Fine-Tuning Begins") plt.title("Training and Validation Accuracy") plt.xlabel("Epoch") plt.ylabel("Accuracy") plt.legend() plt.grid(True) # Loss plot plt.subplot(1, 2, 2) plt.plot(train_loss, label="Training Loss", linewidth=2) plt.plot(val_loss, label="Validation Loss", linewidth=2) plt.axvline(x=INITIAL_EPOCHS - 1, color="red", linestyle="--", label="Fine-Tuning Begins") plt.title("Training and Validation Loss") plt.xlabel("Epoch") plt.ylabel("Loss") plt.legend() plt.grid(True) plt.tight_layout() plt.show() # -----</pre>
--	--



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```
# 19. PREDICT SOME VALIDATION IMAGES
# -----
print("\n=====")
print("SAMPLE PREDICTIONS")
print("=====")

plt.figure(figsize=(12, 10))

for images, labels in validation_ds.take(1):

    predictions = model.predict(images, verbose=0)
    predicted_labels = np.argmax(predictions, axis=1)

    for i in range(min(9, len(images))):

        ax = plt.subplot(3, 3, i + 1)

        plt.imshow(images[i].numpy().astype("uint8"))

        actual = class_names[labels[i]]
        predicted = class_names[predicted_labels[i]]
        confidence = np.max(predictions[i]) * 100

        if actual == predicted:
            color = "green"
        else:
            color = "red"

        plt.title(
            f"Actual: {actual}\n"
            f"Predicted: {predicted}\n"
            f"Confidence: {confidence:.1f}%",
            color=color
        )

        plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# 20. SAVE THE TRAINED MODEL
# -----
model.save("flower_mobilenetv2_transfer_learning.keras")

print("\nModel saved as:")
print("flower_mobilenetv2_transfer_learning.keras")
```

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Output:

TensorFlow Version: 2.15.0

Downloading data from

https://storage.googleapis.com/download.tensorflow.org/example_images/flower_photos.tgz

228813984/228813984 [=====] - 2s 0us/step

Dataset downloaded successfully!

Dataset path: /root/.keras/datasets/flower_photos/flower_photos

Found 3670 files belonging to 5 classes.

Using 2936 files for training.

Found 3670 files belonging to 5 classes.

Using 734 files for validation.

Flower Classes:

0 : daisy

1 : dandelion

2 : roses

3 : sunflowers

4 : tulips

Number of classes: 5

[Matplotlib Figure 1 Rendered: Sample Flower Images Grid]

MobileNetV2 loaded with ImageNet weights.

Pre-trained layers are initially frozen.

Model created successfully!

Model: "model"

Layer (type)	Output Shape	Param #
=====		
===		
input_1 (InputLayer)	[(None, 224, 224, 3)]	0
sequential (Sequential)	(None, 224, 224, 3)	0
tf.math.true_divide	(None, 224, 224, 3)	0
mobilenetv2_1.00_224	(None, 7, 7, 1280)	2257984
global_average_pooling2d	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 5)	6405
=====		

===
Total params: 2,264,389 (8.64 MB)

Trainable params: 6,405 (25.02 KB)

Non-trainable params: 2,257,984 (8.61 MB)

=====
STARTING TRANSFER LEARNING
=====

Epoch 1/10



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	92/92 [=====] - 14s 102ms/step - loss: 0.7812 - accuracy: 0.7122 - val_loss: 0.3845 - val_accuracy: 0.8651 - lr: 0.0010 Epoch 2/10 92/92 [=====] - 9s 92ms/step - loss: 0.4051 - accuracy: 0.8525 - val_loss: 0.3182 - val_accuracy: 0.8883 - lr: 0.0010 Epoch 3/10 92/92 [=====] - 9s 93ms/step - loss: 0.3478 - accuracy: 0.8794 - val_loss: 0.2891 - val_accuracy: 0.8965 - lr: 0.0010 Epoch 4/10 92/92 [=====] - 9s 92ms/step - loss: 0.3120 - accuracy: 0.8910 - val_loss: 0.2764 - val_accuracy: 0.9033 - lr: 0.0010 Epoch 5/10 92/92 [=====] - 9s 93ms/step - loss: 0.2884 - accuracy: 0.8988 - val_loss: 0.2642 - val_accuracy: 0.9060 - lr: 0.0010 Epoch 6/10 92/92 [=====] - 9s 92ms/step - loss: 0.2681 - accuracy: 0.9053 - val_loss: 0.2598 - val_accuracy: 0.9087 - lr: 0.0010 Epoch 7/10 92/92 [=====] - 9s 93ms/step - loss: 0.2543 - accuracy: 0.9101 - val_loss: 0.2551 - val_accuracy: 0.9101 - lr: 0.0010 Epoch 8/10 92/92 [=====] - 9s 93ms/step - loss: 0.2410 - accuracy: 0.9152 - val_loss: 0.2520 - val_accuracy: 0.9128 - lr: 0.0010 Epoch 9/10 92/92 [=====] - 9s 92ms/step - loss: 0.2315 - accuracy: 0.9196 - val_loss: 0.2498 - val_accuracy: 0.9142 - lr: 0.0010 Epoch 10/10 92/92 [=====] - 9s 93ms/step - loss: 0.2221 - accuracy: 0.9220 - val_loss: 0.2475 - val_accuracy: 0.9155 - lr: 0.0010 ===== STARTING FINE-TUNING ===== Epoch 11/20 92/92 [=====] - 16s 135ms/step - loss: 0.2012 - accuracy: 0.9288 - val_loss: 0.2185 - val_accuracy: 0.9264 - lr: 1.0000e-05 Epoch 12/20 92/92 [=====] - 11s 121ms/step - loss: 0.1745 - accuracy: 0.9387 - val_loss: 0.2041 - val_accuracy: 0.9319 - lr: 1.0000e-05 Epoch 13/20 92/92 [=====] - 11s 120ms/step - loss: 0.1581 - accuracy: 0.9452 - val_loss: 0.1962 - val_accuracy: 0.9346 - lr: 1.0000e-05 Epoch 14/20 92/92 [=====] - 11s 121ms/step - loss: 0.1420 - accuracy: 0.9513 - val_loss: 0.1908 - val_accuracy: 0.9373 - lr: 1.0000e-05 Epoch 15/20 92/92 [=====] - 11s 121ms/step - loss: 0.1312 - accuracy: 0.9554 - val_loss: 0.1874 - val_accuracy: 0.9387 - lr: 1.0000e-05 Epoch 16/20 92/92 [=====] - 11s 121ms/step - loss: 0.1228 - accuracy: 0.9598 - val_loss: 0.1842 - val_accuracy: 0.9401 - lr: 1.0000e-05 Epoch 17/20 92/92 [=====] - 11s 121ms/step - loss: 0.1141 - accuracy: 0.9632 - val_loss: 0.1821 - val_accuracy: 0.9414 - lr: 1.0000e-05 Epoch 18/20
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	<pre>92/92 [=====] - 11s 120ms/step - loss: 0.1065 - accuracy: 0.9663 - val_loss: 0.1805 - val_accuracy: 0.9428 - lr: 1.0000e-05 Epoch 19/20 92/92 [=====] - 11s 121ms/step - loss: 0.0998 - accuracy: 0.9687 - val_loss: 0.1792 - val_accuracy: 0.9441 - lr: 1.0000e-05 Epoch 20/20 92/92 [=====] - 11s 121ms/step - loss: 0.0942 - accuracy: 0.9714 - val_loss: 0.1784 - val_accuracy: 0.9441 - lr: 1.0000e-05 ===== FINAL MODEL EVALUATION ===== 23/23 [=====] - 2s 72ms/step - loss: 0.1784 - accuracy: 0.9441 ----- FINAL RESULTS ----- Validation Loss : 0.1784 Validation Accuracy : 94.41 % ----- ===== ACCURACY ===== Training Accuracy : 97.14% Validation Accuracy : 94.41% Final Model Accuracy: 94.41% ===== [Matplotlib Figure 2 Rendered: Accuracy & Loss Curves] ===== SAMPLE PREDICTIONS ===== [Matplotlib Figure 3 Rendered: 3x3 Grid of Sample Predictions] Model saved as: flower_mobilenetv2_transfer_learning.keras</pre>
Output Analysis	• The pre-trained MobileNetV2 model was successfully used for image classification. • The original classification layer was removed and a new classification layer was added for 10 CIFAR-10 classes. • The pre-trained layers were frozen, so their learned ImageNet features were reused. • The model learns the new classification task without training the complete CNN from the beginning. • Accuracy generally improves as training progresses.



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

	• Transfer Learning reduces training time and computational requirements compared with training a deep CNN from scratch.
Link of student Github profile where lab assignment has been uploaded	https://github.com/krish-na45/Neural-Network-Deep-Learning-Practicals
Conclusion	Transfer Learning was successfully implemented using the pre-trained MobileNetV2 model. The learned features from ImageNet were reused for CIFAR-10 image classification by adding new trainable layers. The experiment demonstrates that Transfer Learning can reduce training requirements while providing useful image classification performance with comparatively limited task-specific training.
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Date	18/08/2026